
Algorithm 4 Accumulation Motion Detector (AccMotionDet)

Require: Downsampled frames $\tilde{f}_t, \tilde{f}_{\text{prev}}$; accumulated mask $\mathbf{K}_{t-1}$ (initialized to $\mathbf{0}$)

Require: τ_d : pixel difference sensitivity, ω : motion increment, τ_m : activation threshold, $K_{\max}$: accumulation cap, δ : decay rate

Ensure: $\mathbf{M}_t \in \{0, 255\}$: binary foreground mask; $\mathbf{K}_t$: updated accumulated mask

- 1: Convert $\tilde{f}_t$ and $\tilde{f}_{\text{prev}}$ to grayscale: F_t, F_{prev}
 - 2: $\mathbf{D}_t \leftarrow |F_t - F_{\text{prev}}|$ $\triangleright$ Absolute inter-frame difference
 - 3: $\Delta_t(i, j) \leftarrow \begin{cases} \omega & \mathbf{D}_t(i, j) > \tau_d \\ 0 & \text{otherwise} \end{cases}$ $\triangleright$ Instantaneous increment map
 - 4: $\mathbf{K}_t \leftarrow \min(\mathbf{K}_{t-1} + \Delta_t, K_{\max})$ $\triangleright$ Accumulate and cap
 - 5: $\mathbf{K}_t \leftarrow \max(\mathbf{K}_t - \delta, 0)$ $\triangleright$ Temporal decay
 - 6: $\mathbf{M}_t(i, j) \leftarrow \begin{cases} 255 & \mathbf{K}_t(i, j) \geq \tau_m \\ 0 & \text{otherwise} \end{cases}$ $\triangleright$ Threshold to binary mask
 - 7: **return** $\mathbf{M}_t, \mathbf{K}_t$
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